Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 6

Time Allowed: 30 Minutes

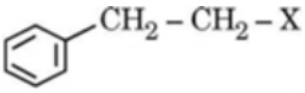
General Instructions:

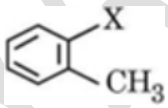
Read the following instructions carefully.

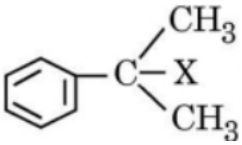
1. There are 16 questions in this question paper with internal choice.
2. **All questions are compulsory.**
3. **Use of log tables and calculators is not allowed.**

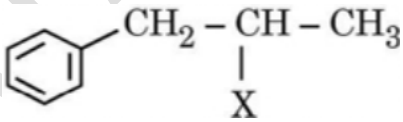
Section A

1. Which of the following is a benzylic halide? **[1]**

a) 

b) 

c) 

d) 

2. **Night-Blindness** is caused by the deficiency of Vitamin: **[1]**

a) C b) K

c) A d) B

3. CH_3 - O - CH_3 when treated with excess HI gives: **[1]**

a) $CH_3 - OH + CH_3 - I$ b) CH_3 - I + CH_4

c) $2CH_3$ - OH d) $2CH_3$ - I

4. Which of the following compounds will undergo self-condensation in the presence of dilute NaOH solution? **[1]**

a) CH_3CH_2CHO b) C_6H_5CHO

c) H-CHO d) $(CH_3)_3C$ - CHO

5. Rate of reaction does not remain constant throughout because **[1]**

a) The density of reactants keep on changing b) Volume of reactants keep on changing

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c) Concentration of reactants keep on changing d) Temperature of reactants keep on changing

6. Match the items of column I with appropriate entries of column II. **[1]**

Column I	Column II
(a) For electrolyte $CaSO_4$	(i) i = 5
(b) For ideal Solution	(ii) i = 2
(c) For electrolyte $Al_2(SO_4)_3$	(iii) i = 4
(d) For electrolyte Na_3PO_4	(iv) i = 1

a) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv). b) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii).

c) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii). d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i).

7. Which of the following belongs to the class of Vinyl halides? **[1]**

a) $CH_2 = CH - CH_2 - Br$ b) $CH_2 = \underset{\underset{Br}{|}}{C} - CH_3$

c) $CH_2 = CHCH_2CH_2Cl$ d) $CH \equiv C - Br$

8. The magnetic nature of elements depends on the presence of unpaired electrons. Identify the configuration of transition element, which shows highest magnetic moment. **[1]**

a) $3d^7$ b) $3d^8$

c) $3d^2$ d) $3d^5$

9. The half-life periods of a reaction at initial concentration of 0.1 mol/L and 0.5 mol/L are 200 s and 40 s respectively. The order of the reaction is **[1]**

a) 0 b) 1

c) 2 d) $\frac{1}{2}$

10. The compound which forms acetaldehyde when heated with dilute NaOH is: **[1]**

a) 1 chloroethane b) 1, 1, 1 trichloroethane

c) 1, 2 dichloroethane d) 1, 1 dichloroethane

11. $(CH_3)_2CH\text{-}O\text{-}CH_3$ reacts with HI to give: **[1]**

a) $(CH_3)_2CH\text{-}I + CH_3\text{-}OH$ b) $(CH_3)_2CH\text{-}OH + CH_3\text{-}I$

c) $(CH_3)_2CH\text{-}I + CH_3\text{-}I$ d) $(CH_3)_2CH\text{-}OH + CH_3\text{-}OH$

12. In order to prepare a 1° amine from an alkyl halide with simultaneous addition of one CH_2 group in the carbon chain, the reagent used as source of nitrogen is _____. **[1]**

a) Potassium phthalimide, $C_6H_4(CO)_2N^-K^+$ b) Ethanolic NaCN

c) Sodium amide, $NaNH_2$ d) Sodium azide, NaN_3

13. **Assertion (A):** Uracil base is present in DNA. **[1**

Reason (R): DNA undergoes self-replication.

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a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion (A):** Carbonyl compounds do not show intermolecular hydrogen bonding. **[1]**

Reason (R): Boiling points of carbonyl compounds are lower than alkanes.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

15. **Assertion (A):** Chlorobenzene is resistant to nucleophilic substitution reaction at room temperature. **[1]**

Reason (R): C–Cl bond gets weaker due to resonance.

a) Both (A) and (R) are true and (R) is the correct explanation of (A).

b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

c) (A) is true, but (R) is false.

d) (A) is false, but (R) is true.

16. **Assertion (A):** Boiling point of butan-1-ol is higher than that of butan-1-amine. **[1]**

Reason (R): Being more polar, butan-1-ol forms stronger intermolecular hydrogen bonds as compared to butan-1-amine

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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